

2026-10-05

Recap

An r -combination of a finite set X is a subset $Y \subseteq X$ such that $N(Y) = r$. The symbol $\binom{n}{r}$ denotes the number of r -combinations of a set of size n .

Theorem 9.5.1

For $n, r \in \mathbb{Z}^{\geq 0}, r \leq n, \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Example 9.5.6

Suppose that 2 people (C, D) do not get along and cannot be in a team together.

$$Z = \{\text{teams of five without (both C and D)}\}$$

$$Y = \{\text{teams of five} - \text{teams of five with both C + D}\}$$

$$N(X) = N(Y) - N(Z), \text{ since } Z \subseteq Y$$

$$N(Y) = \binom{12}{5} \text{ as above}$$

$$N(Z) = \binom{10}{3}$$

$$N(X) = \binom{12}{5} - \binom{10}{3} = \boxed{672}$$

Example 9.5.11

How many ways are there to order the letters in the word *MISSISSIPPI*?

Note: Since I, S, P are repeated, $11!$ overcounts.

Use the recipe:

(1) Choose 4 out of 11 positions for S $\rightarrow \binom{11}{4}$

(2) Choose 4 out of 7 positions for I $\rightarrow \binom{7}{4}$

(3) Choose 2 out of 3 positions for P $\rightarrow \binom{3}{2}$

(4) Choose 1 out of 1 positions for M $\rightarrow \binom{1}{1}$

Number of orderings: $\binom{11}{4} \times \binom{7}{4} \times \binom{3}{2} \times \binom{1}{1} = 34,650$

Theorem 9.5.2.

Suppose a set consists of n objects of k different types and n_i objects are of type i and indistinguishable $\forall i = 1, \dots, k$ and $\sum_{i=1}^k n_i = n$. Then the number of distinguishable permutations of the n objects is

$$\prod_{i=1}^k \binom{n - \sum_{j=1}^{i-1} n_j}{n_i} = \binom{n}{n_1} \times \binom{n - n_1}{n_2} \times \binom{n - n_1 - n_2}{n_3} \times \dots$$

$$\times \binom{n - n_1 - \dots - n_{k-1}}{n_k} = \binom{n!}{n_1! \times n_2! \times \dots \times n_k!}$$

9.6 r-combinations with repetition allowed

Definition

A **multiset** of size r , chosen from a set X of n elements is an unordered selection of elements from X with repetition allowed.

For $X = \{x_1, x_2, \dots, x_n\}$ we write $[x_{i_1}, x_{i_2}, \dots, x_{i_r}]$ for the multiset, where $x_{i_j} \in X, \forall j = 1, \dots, r$ and x_{i_j} may equal each other.

Example 9.6.1.

Multisets of size 3 chosen from $\{1, 2, 3, 4\}$:

[1, 1, 1], [1, 1, 2], [1, 1, 3]
 [1, 1, 4], [1, 2, 2], [1, 2, 4]
 [1, 3, 3], [1, 3, 4], [1, 4, 4]
 [2, 2, 2], [2, 2, 3], [2, 2, 4]
 [2, 3, 3], [2, 3, 4], [2, 4, 4]
 [3, 3, 3], [3, 3, 4], [3, 4, 4]
 [4, 4, 4]

There are 20 multisets of size 3.

To do this systematically, use a table:

Number of ways of distributing the "x" = the number of ways of writing 3 "x" and 3 "", e.g. = number of ways of selecting 3 positions of 6 for the "x", disregarding the order = $\binom{6}{3} = 20$

	1	2	3	4
[1, 1, 1]	xxx			
[1, 1, 2]	xx	x		
[2, 4, 4]		x		xx

Theorem 9.6.1

The number of multisets of size r chosen from n elements is

$$\binom{n+r-1}{r}$$

Example 9.6.2

Select 15 cans of softdrinks from a menu of 5. Each selection corresponds to a multiset of size 15 from a set of 5. Hence, there are

$$\binom{15+5-1}{15} = 3,876 \text{ possibilities}$$

Example 9.6.5

How many sol's are there to $x_1 + x_2 + x_3 + x_4 = 10$ if $x_i \in \mathbb{Z}^+ \forall i = 1, 2, 3, 4$

Think about 4 categories $x_1, \dots, x_4$, from which we select 10 units:

x_1	x_2	x_3	x_4
xxxxxxx	x	x	x
xxxx	xxx	xx	x
x	x	x	xxxxxxx

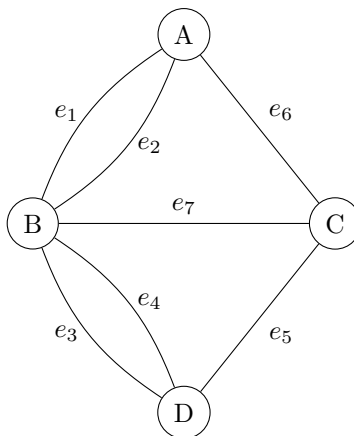
Hence, there are

$$\binom{10 + 4 - 1}{10} = 286 \text{ sol's}$$

Summary of formulas when choosing k elements from n:

	Order matters	Order doesn't matter
Repetition allowed	n^k	$\binom{n+k-1}{k}$
Repetition not allowed	$P(n, k)$	$\binom{n}{k}$

10.1 Trails, Paths and Circuits



Definition (1.4)

A graph G consists of two finite sets and a function:

$V(G) \neq \emptyset$, the set of **vertices**

$E(G)$, the set of **edges**

$\phi_G : E(G) \rightarrow \{\{x, y\} \mid x, y \in V(G)\}$, the **incidence function**

If $\phi_G(e) = \{x, y\}$, e is said to connect x and y

If $\phi_G(e) = \{x\}$, e is a loop.

Two vertices which are connected are called **adjacent** and a vertex not connected to any edge is called **isolated**.

The number of edges connected to a vertex is its degree, $\deg(v)$. A loop is counted twice.